Implementation of a breadth-first search algorithm on a massive-parallel computer

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Abstract

We introduce a fast algorithm of breadth-first search proposed for the Graph 500 benchmark test. We present the result of its implementation on the K-Computer and discuss the optimization of the code.

Keyword: Breadth first search, Graph 500

Processing of graph structures is a major subject in the field of computer science. Because of its importance, a benchmark test called the Graph 500 has recently been proposed [1].

The Graph 500 benchmark test aims to measure the performance of a computer by the calculation speed of breadth-first search of scale-free graphs. The performance highly depends on the speed of the integer operation and the memory bandwidth of the computer. It is also important to consider the efficient use of the interconnect between computation nodes of the computer. This is because recent supercomputers, without exception, have massively parallel architectures with many computation nodes, and the graph processing should be done by these computation nodes communicating each other.

We implemented a fast algorithm of breadth-first search proposed in Ref.[2] on the K Computer at RIKEN. Because the algorithm was devised for the BlueGene computers, we had to take into account the architectural differences between these systems. We discuss the technical problems of the implementation focusing on the optimization of the algorithm considering the structure of the K Computer's interconnect, the TOFU interconnect.

References

[1] http://www.graph500.org/

[2] Checconi et al., Breaking the Speed and Scalability Barriers for Graph Exploration on Distributed-memory Machines, Supercomputing (2012).